Vectors in n Dimensions

Section 1: Vector Arithmetic and Linear Combinations

1 Given $\mathbf{u} = [2, -1, 4]$ and $\mathbf{v} = [-3, 5, 2]$ in $\mathbb{R}^3$, compute $3\mathbf{u} - 2\mathbf{v}$.

a. $[12, -13, 8]$

b. $[0, 7, 16]$

c. $[12, -7, 8]$

d. $[-12, 13, -8]$

2 Is the vector $\mathbf{w} = [7, 4]$ a linear combination of $\mathbf{v}_1 = [1, 2]$ and $\mathbf{v}_2 = [3, -1]$?

a. Yes, $\mathbf{w} = \frac{19}{7}\mathbf{v}_1 + \frac{10}{7}\mathbf{v}_2$

b. No, the vectors are linearly dependent

c. Yes, $\mathbf{w} = 2\mathbf{v}_1 + \mathbf{v}_2$

d. No, because the determinant is zero

3 What is the definition of the span of a set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ in $\mathbb{R}^n$?

a. The set of all possible linear combinations $c_1\mathbf{v}_1 + \ldots + c_k\mathbf{v}_k$ where $c_i \in \mathbb{R}$

b. The set of all vectors perpendicular to $\mathbf{v}_1$

c. The sum of the lengths of the vectors

d. The matrix formed by using the vectors as columns

Section 2: Linear Independence and Dependence

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4 A set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is linearly independent if and only if:

a. $c_1\mathbf{v}_1 + \ldots + c_k\mathbf{v}_k = \mathbf{0}$ implies $c_1 = c_2 = \ldots = c_k = 0$

b. At least one vector is a multiple of another

c. The dot product of any two vectors is zero

d. The number of vectors k is greater than the dimension n

5 Can any set of 4 vectors in $\mathbb{R}^3$ be linearly independent?

a. No, any set of more than n vectors in $\mathbb{R}^n$ is guaranteed to be linearly dependent

b. Yes, provided none of the vectors is the zero vector

c. Yes, if they all have unit length

d. Only if their cross products are non-zero

6 Are the vectors $\mathbf{v}_1 = [1, 2, 0]$, $\mathbf{v}_2 = [0, 1, 3]$, and $\mathbf{v}_3 = [2, 5, 3]$ linearly independent?

a. No, because $\mathbf{v}_3 = 2\mathbf{v}_1 + \mathbf{v}_2$

b. Yes, they span all of $\mathbb{R}^3$

c. No, because they contain a zero entry

d. Yes, because their determinant is 1

Section 3: Dot Product, Norm, and Orthogonality

7 Compute the Euclidean norm (length) of the vector $\mathbf{v} = [2, -3, 6]$ in $\mathbb{R}^3$.

a. 7

b. $\sqrt{11}$

c. 49

d. $\sqrt{41}$

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8 Find the unit vector in the direction of $\mathbf{u} = [-1, 2, 2]$.

a. $\left[-\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right]$

b. $[-1, 2, 2]$

c. $\left[-\frac{1}{9}, \frac{2}{9}, \frac{2}{9}\right]$

d. $\left[-\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}, 0\right]$

9 What does the Cauchy-Schwarz inequality state for any vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$?

a. $|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\|\|\mathbf{v}\|$

b. $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$

c. $\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\|$

d. $|\mathbf{u} \cdot \mathbf{v}| \geq \|\mathbf{u}\|\|\mathbf{v}\|$

Section 4: Orthogonal Projections

10 What is the formula for the orthogonal projection of vector $\mathbf{y}$ onto a non-zero vector $\mathbf{u}$?

a. $\text{proj}_{\mathbf{u}}(\mathbf{y}) = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}}\mathbf{u}$

b. $\text{proj}_{\mathbf{u}}(\mathbf{y}) = \frac{\mathbf{y} \cdot \mathbf{u}}{\|\mathbf{y}\|}\mathbf{u}$

c. $\text{proj}_{\mathbf{u}}(\mathbf{y}) = (\mathbf{y} \cdot \mathbf{u})\mathbf{u}$

d. $\text{proj}_{\mathbf{u}}(\mathbf{y}) = \mathbf{y} - \mathbf{u}$

11 Compute the projection of $\mathbf{y} = [7, 6]$ onto $\mathbf{u} = [4, 2]$.

a. $[8, 4]$

b. $[4, 2]$

c. $[7, 6]$

d. $[2, 1]$

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12 If $\hat{\mathbf{y}} = \text{proj}_{\mathbf{u}}(\mathbf{y})$, what is the geometric relationship between $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$ and $\mathbf{u}$?

a. $\mathbf{z}$ is orthogonal to $\mathbf{u}$

b. $\mathbf{z}$ is parallel to $\mathbf{u}$

c. $\mathbf{z}$ has unit length

d. $\mathbf{z} = \mathbf{0}$ always

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